5G Figures Explained

Screenshot-based revision resume - 20 pages

This version keeps the same 20-page structure but fills the pages with explanation, important notes and common mistakes. The aim is to help you understand why each figure matters, not only what appears in it.

How to use this PDF
Read the explanation first, then the important note. Use the common mistake box as an exam checklist before answering.

Scope
Only the screenshots provided from Chapter 9 Transport and Chapter 10 5G Performance are included. No unrelated course sections are added.

Page structure: cover + figure map + 18 explained screenshots = exactly 20 pages.

Figure map and revision path

Pages 3-9 cover transport: tiers, dimensioning, port speed, RTT, NG connectivity and slicing. Pages 10-20 cover performance: peak rate, latency, TDD, MEC, link budget and antenna gain.

1. Figure 9.1 - Mobile backhaul network tiers - Chapter 9 - 5G Transport / Mobile backhaul tiers
2. Figure 9.3 - Dimensioning example gNB configuration with three sectors - Chapter 9 - Access link dimensioning
3. Figure 9.4 - Backhaul capacity NG/F1 requirement - Chapter 9 - Backhaul and high-layer fronthaul capacity
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14. Figure 10.12 - TDD frame adds latency compared to FDD - Chapter 10 - TDD latency
15. Figure 10.17 - Distributed architecture with mobile edge computing - Chapter 10 - Low-latency architecture / MEC
16. Table 10.13 - Downlink link budget for sub-6-GHz TDD - Chapter 10 - Downlink link budget
17. Table 10.12 - Uplink link budget for sub-6-GHz TDD - Chapter 10 - Uplink link budget
18. Figure 10.22 - Antenna gain and size benchmarking - Chapter 10 - Beamforming antenna gain

Priority if you have little time

First revise Figure 9.4 and Figure 9.7 for transport capacity, Tables 10.8/10.9 for latency, Tables 10.12/10.13 for link budget, and Figure 10.22 for massive MIMO gain.

Figure 9.1 - Mobile backhaul network tiers

Chapter 9 - 5G Transport / Mobile backhaul tiers

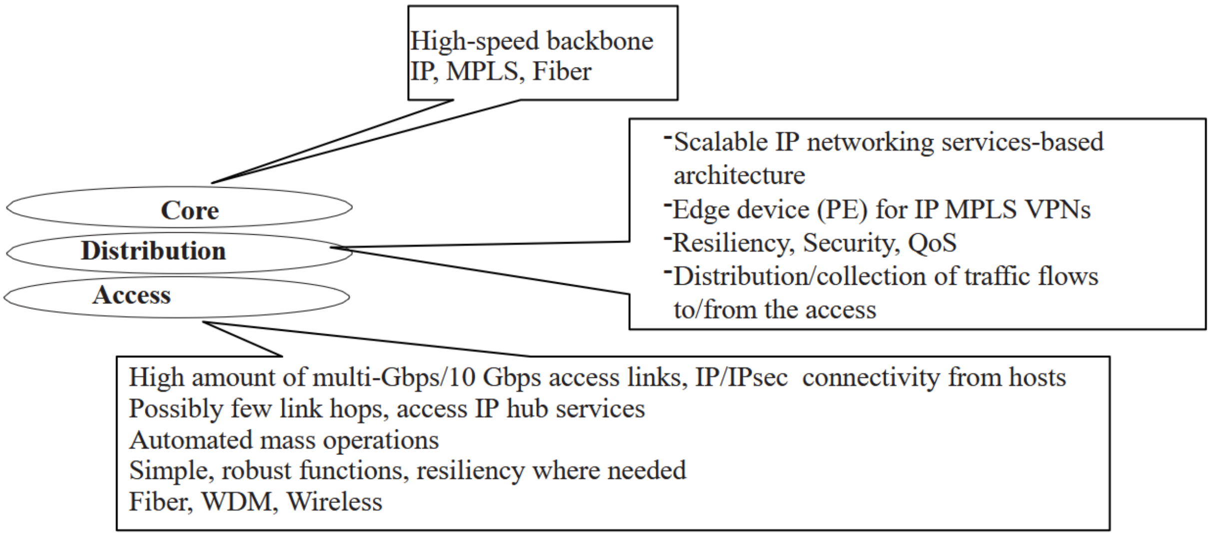


Figure 9.1 Mobile backhaul network tiers.

Screenshot 1. Figure 9.1 - Mobile backhaul network tiers.

Explanation

The important idea is that a 5G transport network is not designed as one uniform pipe. It is split into access, distribution and core because each part has a different engineering problem. The access tier is close to the gNBs, so it contains many links and must be cheap, simple, highly automated and robust. The distribution tier aggregates traffic and is where IP/MPLS VPN services, QoS, security and resiliency are usually organized. The core tier is the high-speed backbone that carries already aggregated traffic toward core network functions and data centers. For revision, read the arrows as a cost and aggregation gradient: many small but capacity-hungry access links feed fewer, stronger distribution/core links. Low-latency use cases change the usual hierarchy because functions such as UPF or MEC must move closer to access to avoid long propagation delay.

Important notes

Remember the role of each tier: access = first mile and massive number of links; distribution = aggregation plus IP/MPLS services; core = national or regional high-speed backbone. 5G often forces access upgrades because radio capacity increases faster than old transport links.

Exam answer pattern

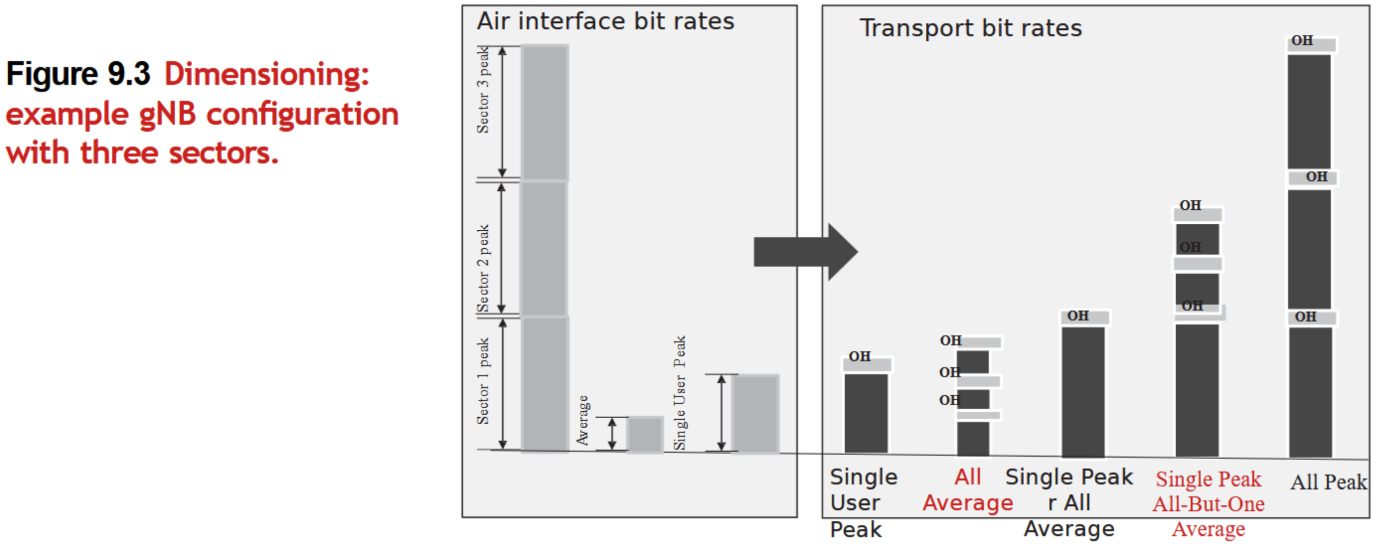
In an exam answer, start by defining the three tiers, then explain why access is cost-sensitive, distribution is service/QoS-oriented and core is high-capacity. Finish by linking MEC/UPF placement to low-latency needs.

Common mistake

Do not say that access/distribution/core are only physical distances. They are design zones with different capacity, cost, QoS, security and resiliency constraints.

Figure 9.3 - Dimensioning example gNB configuration with three sectors

Chapter 9 - Access link dimensioning



Screenshot 2. Figure 9.3 - Dimensioning example gNB configuration with three sectors.

Explanation

This figure gives the logic for estimating the required transport capacity when exact traffic demand is unknown. Each sector has a peak radio rate, but real networks rarely have all sectors at peak simultaneously. The engineer therefore chooses a dimensioning scenario: all sectors at average load, one sector at peak, one sector at peak while the others are average, or all sectors at peak. The small OH blocks represent protocol overhead, meaning the transport bit rate must be larger than the radio payload rate. The figure is useful because it separates user experience from site capacity: the single-user peak protects the maximum speed a user can observe, while sector-based dimensioning protects the full site capacity. In exam answers, justify which case is chosen instead of only giving a number.

Important notes

A safe method is: estimate radio rates, select a dimensioning scenario, add overhead, then choose the next standard port speed above the result. Single-user peak is a minimum user-experience requirement.

Exam answer pattern

Answer method: choose the dimensioning scenario, calculate the required rate, add protocol overhead, then justify the selected transport capacity. State whether you are protecting user peak or sector/site capacity.

Common mistake

A common mistake is sizing only from average traffic. That may pass under normal load but fails when one sector becomes hot or when users expect peak-rate service.

Figure 9.4 - Backhaul capacity NG/F1 requirement

Chapter 9 - Backhaul and high-layer fronthaul capacity

Air Interface						Transmission bit rates	
Frequency	Bandwidth	SU-MIMO streams	MU-MIMO streams	Sector Peak rate	Single User Peak rate	Alternative A: Air if based	Alternative B: Single user peak rate based
>60 GHz	5 MHz TDD	2	4	2	4	2 sector config. F1: 58 Gbps NG: 59 Gbps	F1: 29 Gbps NG: 30 Gbps
24-39 GHz	800 MHz	2	4	2	4	2 sector config. F1: 14 Gbps NG: 14 Gbps	F1: 6.9 Gbps NG: 7.1 Gbps
3.3-4.8 GHz	100 MHz	4	8	4	8	3 sector config. F1: 4.6 Gbps NG: 4.7Gbps	F1: 2.3 Gbps NG: 2.4 Gbps
<1 GHz	20 MHz	2	2	2	2	3 sector config. F1: 0.29 Gbps NG: 0.3Gbps	F1: 0.29 Gbps NG:0.3 Gbps

Figure 9.4 Backhaul capacity (NG, F1) requirement. 24/61

Screenshot 3. Figure 9.4 - Backhaul capacity NG/F1 requirement.

Explanation

This figure connects the radio interface to the transport interface. The left part gives radio assumptions such as spectrum, bandwidth and MIMO streams; these create sector peak and single-user peak rates. The right part converts those radio rates into F1 and NG transport bit rates after overhead. Alternative A is based on air-interface capacity, so it protects the site or sector peak and normally gives larger requirements. Alternative B is based on the single-user peak rate, so it ensures one user can experience the advertised peak but may not protect the full cell capacity. For the 3.3-4.8 GHz, 100 MHz, three-sector case, requirements are already several Gbps, so 1G is not acceptable and 10G becomes the practical class. The message is that 5G capacity gains create immediate backhaul pressure.

Important notes

Know the difference between F1 and NG: F1 links DU to CU in split RAN; NG links RAN/CU/gNB to the 5G Core. Both may need multi-Gbps capacity when radio bandwidth is large.

Exam answer pattern

Answer method: derive air-interface peak from bandwidth and streams, add overhead, compare Alternative A and B, then select a transport class. Explain why 1G is insufficient for mid-band 5G sites.

Common mistake

Do not confuse MHz of radio bandwidth with Gbps of transport capacity. Bandwidth, MIMO streams, DL/UL ratio, modulation and overhead all influence the final transport rate.

Figure 9.7 - Ethernet port speeds and 5G requirements

Chapter 9 - Client ports

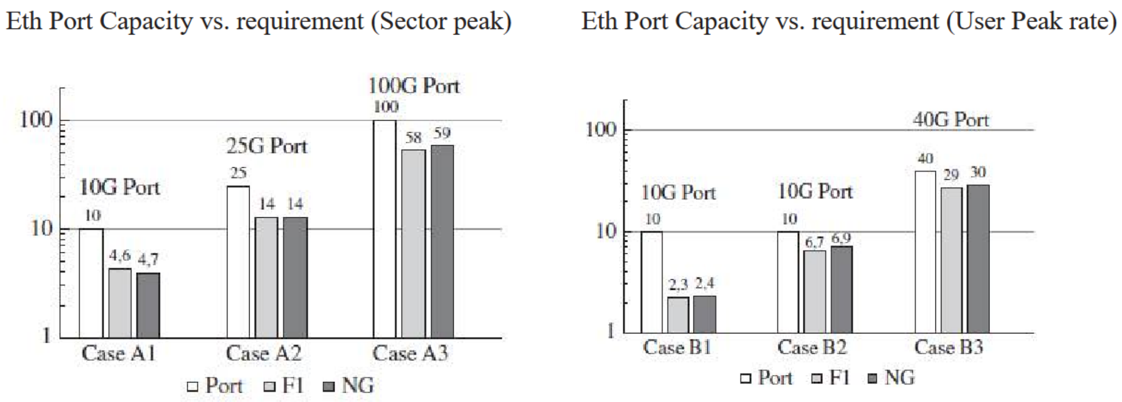


Figure 9.7 Ethernet port speeds and 5G requirements.

Screenshot 4. Figure 9.7 - Ethernet port speeds and 5G requirements.

Explanation

This figure translates calculated capacity into a concrete equipment decision. Once the F1 or NG requirement is known, the network planner must select a standard Ethernet port: 10G, 25G, 40G or 100G. The left chart uses sector-peak dimensioning, so it is stricter; the right chart uses single-user peak dimensioning, so it can lead to smaller ports. The exam method is simple: compare requirement with available port classes and choose the next port above the requirement. For example, about 4.6-4.7 Gbps needs a 10G port, about 14 Gbps needs 25G, about 29-30 Gbps needs 40G, and about 58-59 Gbps needs 100G. The figure also reminds you that the port must leave room for protocol overhead, control traffic, synchronization, growth and statistical bursts.

Important notes

Port selection is a step function: requirements do not need to equal the port speed. You choose the first standard port that safely exceeds the calculated need.

Exam answer pattern

Answer method: take the required F1/NG bit rate and round up to the next Ethernet class. Mention that port choice must include overhead, burst traffic, future growth and non-user-plane traffic.

Common mistake

Do not choose a 1G or exactly equal-capacity port just because the average traffic is lower. Congestion appears quickly when utilization approaches high levels.

Figure 9.6 - RTT components

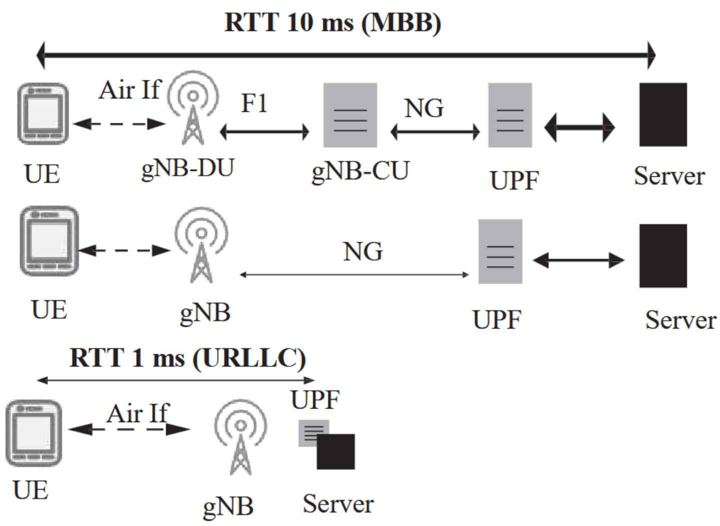


Figure 9.6 RTT components.

Screenshot 5. Figure 9.6 - RTT components.

Explanation

This figure explains that RTT is an end-to-end sum, not just radio latency. The packet travels through UE processing, the air interface, gNB-DU, F1, gNB-CU, NG, UPF, server path and then returns. For MBB, a 10 ms target can tolerate several separated components as long as transport and processing remain controlled. For URLLC, a 1 ms target is so strict that the architecture must collapse: the gNB, UPF and server have to be close, often at an edge site, because long F1/NG/core paths consume the budget. The figure is therefore a design argument: low latency requires both short radio timing and short physical/topological distance. You should mention propagation delay, processing delay and interface delay together.

Important notes

RTT = sum of radio transmission, alignment, processing, transport, core and server path delays. URLLC requires local breakout/MEC because distance alone can break the latency budget.

Exam answer pattern

Answer method: write RTT as a sum of components. For URLLC, explain why gNB, UPF and server must be physically close; for MBB, explain that a few ms of transport can still fit in a 10 ms budget.

Common mistake

Do not answer latency questions only with TTI or subcarrier spacing. That explains radio latency, but not complete end-to-end RTT.

Figure 9.9 - NG interface example (classical gNB)

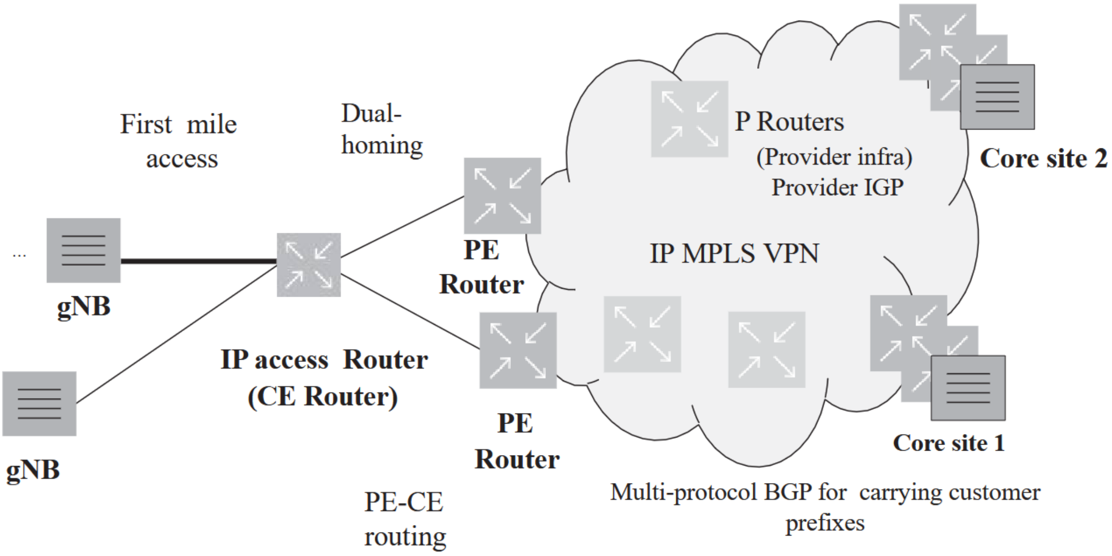


Figure 9.9 NG interface example (classical gNB).

Screenshot 6. Figure 9.9 - NG interface example (classical gNB).

Explanation

This figure explains how a classical gNB reaches the 5G Core through an IP/MPLS VPN transport service. The gNB uses a first-mile access link toward a CE router, then connects to PE routers that attach the customer VPN to the provider network. P routers in the middle carry MPLS/IP traffic across the provider infrastructure; they do not need to know every customer route. MP-BGP distributes VPN routes between PE routers, enabling each customer or service to stay logically separated. Dual-homing adds resiliency by connecting to two PE routers, so a failure on one path does not isolate the gNB. In exam terms, the NG interface is logical IP connectivity between RAN and Core; the physical path can be complex as long as reachability, QoS, security and resiliency are respected.

Important notes

Key vocabulary: CE = customer edge, PE = provider edge, P = provider core router, MP-BGP = route distribution for VPNs. NG-U carries user plane; NG-C carries control plane.

Exam answer pattern

Answer method: identify CE, PE, P routers, MP-BGP and VPN separation. Then explain how dual-homing improves resiliency and how NG remains a logical IP interface over the transport network.

Common mistake

Do not interpret the drawing as a direct cable from gNB to UPF. It is an IP transport service providing logical connectivity over a shared provider network.

Figure 9.10 - Network slicing in transport example

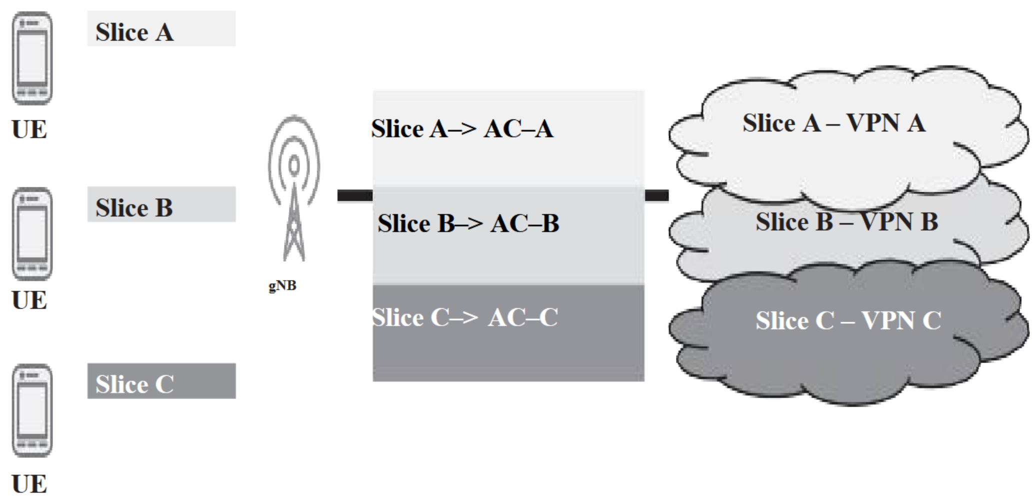


Figure 9.10 Network slicing in transport example.

Screenshot 7. Figure 9.10 - Network slicing in transport example.

Explanation

This figure shows the transport meaning of network slicing: slices are not only radio/core labels; they may require separated transport services. A UE associated with Slice A is mapped by the gNB toward attachment circuit AC-A and then to VPN A. Slice B and Slice C are mapped similarly to their own attachment circuits and VPNs. The goal is logical isolation: each slice can have its own routing table, QoS, security policy, capacity guarantee or local breakout. This is especially important when customers or services have different requirements, such as industrial URLLC, public safety, enterprise VPN or best-effort broadband. For exams, explain that transport slicing can be implemented with IP/MPLS VPN or VRF separation, but not every slice always needs a physically separate network.

Important notes

AC means attachment circuit. VRF/VPN separation allows different slices to share infrastructure while keeping routes and service policies isolated.

Exam answer pattern

In an exam answer, start by defining the three tiers, then explain why access is cost-sensitive, distribution is service/QoS-oriented and core is high-capacity. Finish by linking MEC/UPF placement to low-latency needs.

Common mistake

Do not assume slicing always means separate fiber or separate hardware. Most slicing is logical separation over shared infrastructure, unless strict availability/security requires more.

Figure 10.1 - Approximate peak data rates

Chapter 10 - Peak data rates

Spectrum	Bandwidth	MIMO		Peak rate
>50 GHz	5 GHz TDD	2x2/4x4	➡	50/100 Gbps
24–39 GHz	800 MHz TDD	2x2/4x4	➡	8/16 Gbps
2.5–5.0 GHz	100 MHz TDD	4x4	➡	2 Gbps
<2.7 GHz	20 MHz FDD	4x4	➡	0.4 Gbps
<1 GHz	10 MHz FDD	2x2	➡	0.1 Gbps

Screenshot 8. Figure 10.1 - Approximate peak data rates.

Explanation

This figure gives the intuition behind 5G peak-rate evolution: peak rate grows mainly with bandwidth and MIMO capability. Low bands below 1 GHz have only about 10 MHz FDD and 2x2 MIMO, so the peak is around 0.1 Gbps. Mid-bands around 2.5-5 GHz can use around 100 MHz TDD and 4x4 MIMO, reaching around 2 Gbps. Millimeter-wave bands have much wider bandwidth, around 800 MHz, so they can reach 8-16 Gbps depending on MIMO assumptions. The extreme future case above 50 GHz assumes several GHz of bandwidth and can reach tens of Gbps. The exam conclusion is that low-band coverage is good but peak rate is limited, while high-band capacity is excellent but coverage and propagation are harder.

Important notes

Peak rate increases with three levers: bandwidth, number of spatial streams and bits per symbol. Frequency itself does not magically create data rate; it enables wider usable bandwidth and smaller antennas.

Exam answer pattern

Answer method: compare bands through bandwidth and MIMO. State the trade-off: low bands give coverage with limited rate, mid-bands balance rate and coverage, mmWave gives high capacity but short range.

Common mistake

Do not say high frequency automatically means high speed. The speed comes from available bandwidth, MIMO streams and modulation, while high frequency also increases propagation loss.

Data rate formula - bandwidth x streams x bits per symbol

Chapter 10 - Peak data-rate formula

$$Datarate = Bandwidth \ \ x \ \ (Number \ \ of \ \ (Antenna \ \ streams)) \ \ x \ \ (\frac{bits}{symbol} \, per \, stream)$$

Screenshot 9. Data rate formula - bandwidth x streams x bits per symbol.

Explanation

This formula is the simplified peak-rate model. Bandwidth tells how many symbols can be transmitted per second; the number of antenna streams tells how many independent spatial layers are sent in parallel; bits per symbol comes from modulation order, for example QPSK = 2 bits/symbol, 64-QAM = 6 bits/symbol and 256-QAM = 8 bits/symbol. The formula explains the main direction of change, but it is not the final practical throughput because real systems lose resources to guard bands, pilots, PDCCH, DMRS, coding overhead and the DL/UL split in TDD. A small example: 100 MHz with 4 streams and 8 bits/symbol gives a theoretical 3.2 Gbps before overhead; the practical value is lower, around the 2 Gbps order in the course assumptions.

Important notes

Use this formula for intuition and rough peak-rate reasoning. For practical peak rates, apply overhead, coding rate, PRB usage and TDD ratio.

Exam answer pattern

Answer method: use the formula for a quick theoretical peak, then explicitly subtract practical losses such as coding, control channels, reference signals, guard bands and TDD DL/UL sharing.

Common mistake

Do not forget that Mbps/Gbps values in tables are after several reductions. The simple formula is an upper-bound intuition, not a complete 5G throughput calculator.

Table 10.4 - Peak data rate calculations

Chapter 10 - Detailed peak-rate calculations

Table 10.4 Peak data rate calculations.

	Sub-1-GHz FDD	2.5–5.0 GHz TDD	24–39 GHz TDD
Bandwidth	10 MHz	100 MHz	800 MHz (2 × 400 MHz)
Subcarrier spacing	15 kHz	30 kHz	120 kHz
Number of PRBs	52	273	528
Modulation	Downlink 256-QAM Uplink 64-QAM	Downlink 256-QAM Uplink 64-QAM	Downlink 64-QAM Uplink 64-QAM
MIMO	Downlink 2 × 2 Uplink no MIMO	Downlink 4 × 4 Uplink no MIMO	Downlink 2 × 2 Uplink no MIMO
PDCCH symbols	1 symbol for full bandwidth	1 symbol using 50% bandwidth	1 symbol using 50% Bandwidth
DMRS symbols SSB	1	1	1
TDD radio	Not considered Downlink 1.0 (FDD) Uplink 1.0 (FDD)	Not considered Downlink 0.74 Uplink 0.23	Not considered Downlink 0.74 Uplink 0.23
Coding rate	0.93	0.93	0.93
Peak rate downlink	**111 Mbps**	**1.80 Gbps**	**5.2 Gbps**
Peak rate uplink	**42 Mbps**	**0.104 Gbps**	**0.80 Gbps**

PRB, Physical Resource Block; MIMO, Multiple Input Multiple Output; PDCCH, Physical Downlink Control Channel; SSB, Synchronization Signal Block.

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Screenshot 10. Table 10.4 - Peak data rate calculations.

Explanation

This table is the practical version of the peak-rate formula. It compares three deployments: sub-1 GHz FDD, mid-band TDD and mmWave TDD. The final rates are not obtained from bandwidth alone; they also depend on PRBs, subcarrier spacing, modulation, MIMO layers, PDCCH overhead, DMRS, TDD split and coding rate. The mid-band case reaches around 1.80 Gbps downlink because it combines 100 MHz, 30 kHz spacing, 4x4 downlink MIMO and high-order 256-QAM, but then loses capacity to control channels and TDD sharing. The mmWave case reaches higher downlink rate because 800 MHz is enormous, even though modulation is only 64-QAM and MIMO is 2x2. The uplink rates are much smaller because uplink has no MIMO in these assumptions and only 23% TDD share.

Important notes

When explaining a peak-rate table, list the factors in order: bandwidth/PRBs, modulation, MIMO streams, overhead symbols, TDD ratio and coding rate.

Exam answer pattern

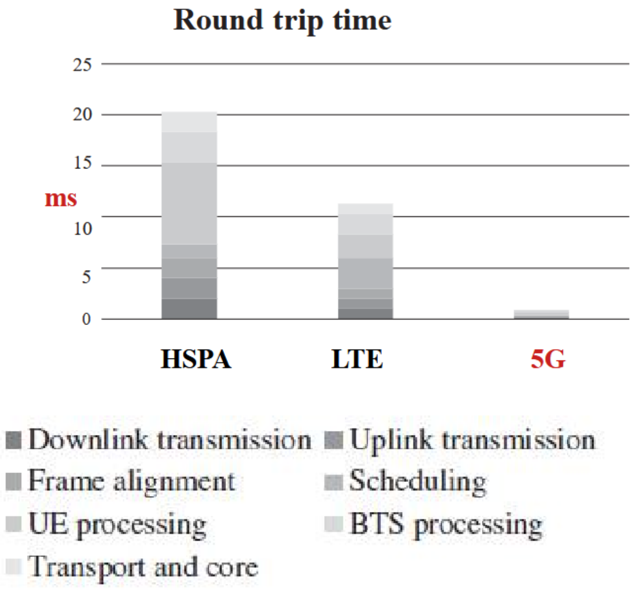
Answer method: explain each row as a practical reduction from the simple formula. The final rates require PRBs, modulation, MIMO, overhead, TDD ratio and coding rate, not just bandwidth.

Common mistake

Do not compare only bandwidth. A smaller bandwidth with better MIMO/modulation may perform well, and a very wide bandwidth still loses capacity to overhead and TDD allocation.

Figure 10.8 - Round trip time evolution

Chapter 10 - RTT evolution



Screenshot 11. Figure 10.8 - Round trip time evolution.

Explanation

This figure explains why 5G reduces latency compared with HSPA and LTE. The gain comes from shorter transmission intervals, shorter frame alignment, faster UE/BTS processing, less scheduling delay and local transport/core assumptions. HSPA is dominated by long transmission and processing delays, giving around 20 ms RTT. LTE improves the radio structure but still has scheduling and processing delays, giving around 11 ms in favorable cases. 5G can go below a few milliseconds because it supports shorter TTI, mini-slots, faster processing and local content. The stacked bars are important: latency is additive, so reducing only one layer is not enough. The figure supports the idea that 5G latency is a system-level improvement, not one isolated feature.

Important notes

Main latency components: downlink/uplink transmission, frame alignment, scheduling, UE processing, BTS processing, transport and core. The biggest improvement is achieved when several components shrink together.

Exam answer pattern

Answer method: compare HSPA, LTE and 5G by identifying which latency components shrink. Avoid saying only "5G is faster"; name TTI, scheduling, processing and local core/transport.

Common mistake

Do not read 5G low latency as guaranteed in every deployment. Centralized core, TDD waiting and retransmissions can raise the real measured RTT.

Table 10.8 - Two-way latency with different subcarrier spacing

Chapter 10 - Full TTI latency and numerology

Delay component	15 kHz	30 kHz	60 kHz	120 kHz
Downlink transmission (ms)	1.0	0.5	0.25	0.125
Uplink transmission (ms)	1.0	0.5	0.25	0.125
Downlink frame alignment (ms)	0.5	0.25	0.125	0.0625
Uplink frame alignment (ms)	0.5	0.25	0.125	0.0625
UE processing – reception (ms)	0.57	0.36	0.30	0.17
UE processing –transmission (ms)	0.71	0.43	0.41	0.32
BTS processing – reception (ms)	0.29	0.18	0.15	0.09
BTS processing – transmission (ms)	0.36	0.22	0.22	0.16
Transport and core (ms)	0.10	0.10	0.10	0.10
Total (ms)	**5.0**	**2.8**	**1.9**	**1.2**

Screenshot 12. Table 10.8 - Two-way latency with different subcarrier spacing.

Explanation

This table shows how numerology affects latency. As subcarrier spacing increases from 15 to 120 kHz, slot duration becomes shorter, so downlink transmission, uplink transmission and frame-alignment time decrease. That is why total RTT falls from about 5.0 ms at 15 kHz to about 1.2 ms at 120 kHz under the table assumptions. However, not all components scale in the same way: UE processing, BTS processing and transport/core delay remain significant. This is why high subcarrier spacing is helpful but not sufficient alone. The table is also an exam calculation template: add every delay component to obtain total RTT, and identify which components change when the subcarrier spacing changes.

Important notes

Higher subcarrier spacing -> shorter slot -> lower radio transmission and alignment delay. But processing and transport/core delays remain part of the RTT budget.

Exam answer pattern

Answer method: add the delay rows and explain why higher subcarrier spacing shortens slot duration. Then state the limitation: transport/core and processing still remain.

Common mistake

Do not claim that 120 kHz is always better. It improves timing but is mostly used in higher bands and can have coverage/propagation trade-offs.

Table 10.9 - Two-way latency with mini-slot transmission

Chapter 10 - Mini-slot latency

Table 10.9 Two-way latency with mini-slot transmission with 15 kHz with UE capability 2.

Delay component	14-symbol	7-symbol	4-symbol	2-symbol
Downlink transmission (ms)	1.0	0.5	0.29	0.14
Uplink transmission (ms)	1.0	0.5	0.29	0.14
Downlink frame alignment (ms)	0.5	0.25	0.14	0.07
Uplink frame alignment (ms)	0.5	0.25	0.14	0.07
UE processing – reception (ms)	0.21	0.21	0.21	0.21
UE processing – transmission (ms)	0.36	0.36	0.36	0.36
BTS processing – reception (ms)	0.11	0.11	0.11	0.11
BTS processing – transmission (ms)	0.18	0.18	0.18	0.18
Transport and core (ms)	0.10	0.10	0.10	0.10
Total (ms)	**4.0**	**2.5**	**1.8**	**1.4**

Screenshot 13. Table 10.9 - Two-way latency with mini-slot transmission.

Explanation

This table explains the specific benefit of mini-slots. With 15 kHz spacing, a full 14-symbol slot gives about 4.0 ms two-way latency in the assumptions shown. When the transmission uses only 7, 4 or 2 symbols, both the transmission time and average frame-alignment waiting time decrease, so total latency drops to 2.5 ms, 1.8 ms and 1.4 ms. Notice the fixed part of the budget: UE processing, BTS processing and transport/core stay the same. Mini-slot transmission therefore mainly reduces the waiting/transmission portion, not every delay source. It is most useful for small urgent packets, such as URLLC payloads, because a large file would require many mini-slots and create too much control overhead and fragmentation.

Important notes

Mini-slot = 2, 4 or 7 OFDM symbols instead of a full 14-symbol slot. It is a latency tool for short packets and urgent traffic.

Exam answer pattern

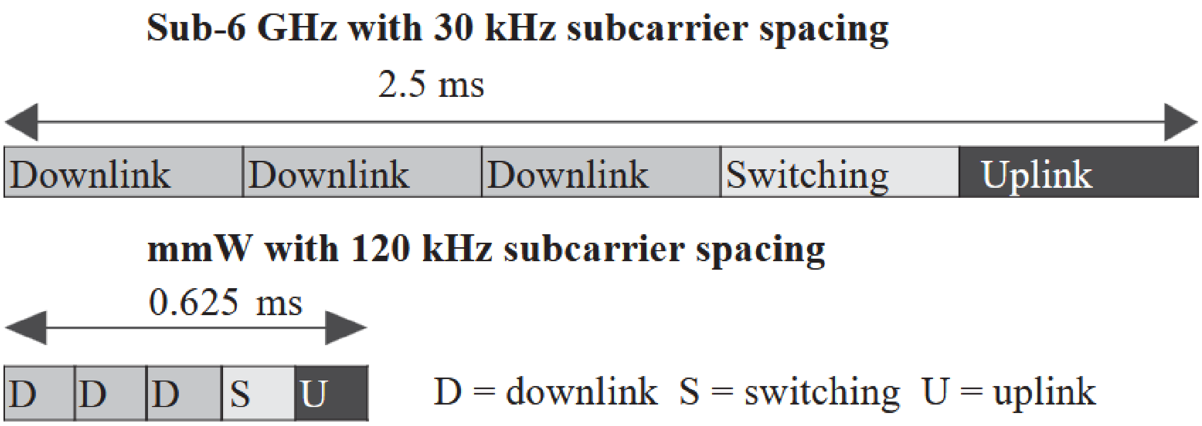
Answer method: compare 14, 7, 4 and 2 symbols. Explain that mini-slots reduce transmission/alignment delay but mainly benefit small urgent packets, not large sustained downloads.

Common mistake

Do not use mini-slots blindly for large eMBB traffic. The PDCCH overhead and fragmentation can reduce efficiency.

Figure 10.12 - TDD frame adds latency compared to FDD

Chapter 10 - TDD latency



Screenshot 14. Figure 10.12 - TDD frame adds latency compared to FDD.

Explanation

This figure explains the waiting penalty created by TDD. In FDD, uplink and downlink are available on separate frequencies, so a packet can be sent in the needed direction without waiting for the frame to switch. In TDD, the same frequency alternates between D, S and U parts. If an uplink packet arrives just after the uplink part has passed, it waits almost an entire TDD frame; on average it waits about half the frame. For sub-6 GHz with 30 kHz spacing, the 2.5 ms frame can add around 1 ms compared with FDD. For mmWave with 120 kHz spacing, the TDD frame is only 0.625 ms, so the added waiting is much smaller. The exam conclusion is that sub-6 GHz TDD is excellent for capacity but not ideal for 1-2 ms RTT targets.

Important notes

Average TDD waiting time is approximately half the TDD frame for the direction that is currently unavailable. Shorter frames reduce this penalty.

Exam answer pattern

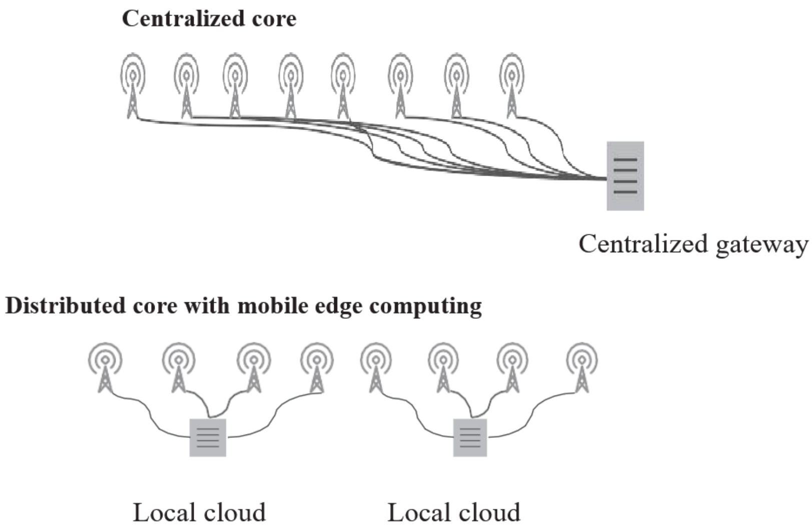
Answer method: compare bands through bandwidth and MIMO. State the trade-off: low bands give coverage with limited rate, mid-bands balance rate and coverage, mmWave gives high capacity but short range.

Common mistake

Do not forget the switching part S. TDD is not simply DL plus UL; switching/guard time is necessary and consumes time resources.

Figure 10.17 - Distributed architecture with mobile edge computing

Chapter 10 - Low-latency architecture / MEC



Screenshot 15. Figure 10.17 - Distributed architecture with mobile edge computing.

Explanation

This figure explains why low latency requires architectural decentralization. In a centralized core, traffic from many radio sites is routed to a distant centralized gateway, even if the application could be local. The distance adds propagation delay and can also increase transport load. In a distributed core with MEC, local clouds or local gateways are placed closer to radio sites, allowing traffic to break out near the user. This shortens the physical path and reduces RTT. The key reasoning is causal: radio latency improvements make transport distance visible; once the air interface becomes fast, the remaining milliseconds are often in fiber distance, gateway placement and server location. MEC is therefore not just cloud computing; it is a placement strategy for latency-sensitive services.

Important notes

MEC/local UPF is essential for services such as URLLC, industrial control, gaming, AR/VR and local enterprise services. It reduces both propagation delay and unnecessary core transport.

Exam answer pattern

Answer method: compare bands through bandwidth and MIMO. State the trade-off: low bands give coverage with limited rate, mid-bands balance rate and coverage, mmWave gives high capacity but short range.

Common mistake

Do not say MEC improves latency only because servers are faster. The main gain here is shorter route and local breakout.

Table 10.13 - Downlink link budget for sub-6-GHz TDD

Chapter 10 - Downlink link budget

Table 10.13 Downlink link budget for sub-6-GHz TDD.

System parameter	PDSCH 50 (Mbps)	PDSCH 10 (Mbps)	PDCCH	PBCH
Data rate (Mbps)	50.0	10.0		
Tx power (W)	200	200	200	200
Tx power (dBm)	**53.0**	**53.0**	**53.0**	**53.0**
Subcarrier spacing (kHz)	30	30	30	30
Total PRBs	275	275	275	275
Used PRBs	138	138	24	20
Downlink share	80 %	80 %		
Antenna gain (dBi)	**25**	**25**	**25**	**22**
Interference margin (dB)	0	0	0	0
Noise figure (dB)	7	7	7	7
SNR (dB)	1.0	−7.3	−6.0	−10.0
Sensitivity (dBm)	**−89.0**	**−97.3**	**−103.6**	**−108.4**
Max path loss (dB)	139.0	147.3	146.1	150.1
Max path loss with beams (dB)	164.0	172.3	171.1	172.1

Screenshot 16. Table 10.13 - Downlink link budget for sub-6-GHz TDD.

Explanation

This table explains how downlink coverage is derived. The transmitter power is 200 W, equal to 53 dBm. The receiver can decode only if received power is above sensitivity. Sensitivity depends on noise bandwidth, noise figure and required SNR: a channel requiring lower SNR or fewer resources can be decoded at a weaker signal. Max path loss is the maximum attenuation between transmitter and receiver that still leaves enough received power. Beamforming then adds antenna gain, so max path loss with beams becomes much larger. Compare PDSCH 50 Mbps and PDSCH 10 Mbps: the lower-rate case has better sensitivity and therefore a higher path-loss budget. PDCCH and PBCH use different PRBs and SNR assumptions; PBCH also has less antenna gain because common/broadcast channels do not use the same user-specific beamforming gain.

Important notes

Exam method: convert power to dBm if needed, calculate sensitivity, calculate max path loss, then add antenna/beam gain. Lower required data rate generally improves coverage.

Exam answer pattern

Answer method: compare bands through bandwidth and MIMO. State the trade-off: low bands give coverage with limited rate, mid-bands balance rate and coverage, mmWave gives high capacity but short range.

Common mistake

Do not add antenna gain twice. The row "max path loss" is before beam gain; "max path loss with beams" already includes it.

Table 10.12 - Uplink link budget for sub-6-GHz TDD

Chapter 10 - Uplink link budget

System parameter	PUSCH 10 (Mbps)	PUSCH 1 (Mbps)	PUSCH 0.1 (Mbps)	Long PUCCH 1 (ACK)	PRACH format 1	SRS
Data rate (Mbps)	10.0	1.0	0.1			
Tx power (W)						
Tx power (dBm)	**26**	**26**	**26**	**26**	**26**	**26**
Subcarrier spacing (kHz)	30	30	30	30	30	30
PRBs	50	5	2	1	3	4
Uplink share	20 %	20 %	20 %			
Antenna gain (dBi)	**25**	**25**	**25**	**25**	**25**	**16**
Interference margin (dB)	0	0	0	0	0	0
Noise figure (dB)	2	2	2	2	2	2
SNR (dB)	4.7	4.7	−2.0	−6.0	−8.0	−10.0
Sensitivity (dBm)	**−94.7**	**−104.7**	**−115.4**	**−122.4**	**−119.7**	**−120.4**
Max path loss (dB)	**120.7**	**130.7**	**141.4**	**148.4**	**145.7**	**146.4**
Max path loss with beams (dB)	**145.7**	**155.7**	**166.4**	**173.4**	**170.7**	**162.4**

Screenshot 17. Table 10.12 - Uplink link budget for sub-6-GHz TDD.

Explanation

This table explains why uplink is often the limiting direction in mid-band 5G. The UE transmits only 26 dBm, far less than the base station downlink power, so the uplink budget starts weaker. PUSCH at 10 Mbps has the lowest max path loss because a higher data rate requires more effective received quality. When the target rate decreases to 1 Mbps or 0.1 Mbps, sensitivity improves and the allowed path loss increases. Control channels such as PUCCH and PRACH have relatively good coverage because their payloads are smaller and SNR requirements are lower. SRS is special: it supports sounding and beamforming but cannot fully benefit from beamforming gain, so it can become a coverage limitation. This table is the main reason for solutions such as low-band uplink, dual connectivity and supplemental uplink.

Important notes

PUSCH is usually the coverage bottleneck for uplink data. Lower uplink rate means better coverage because required SNR and/or bandwidth pressure is lower.

Exam answer pattern

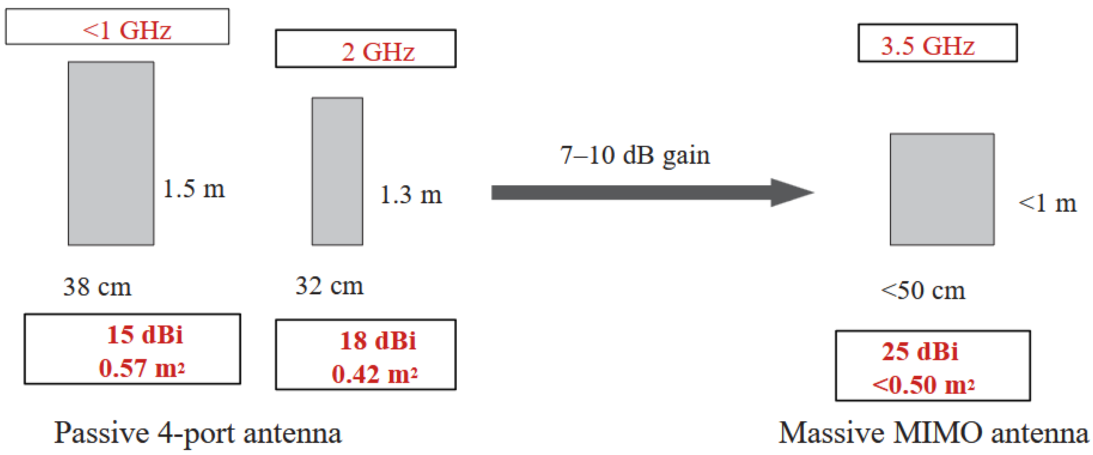
Answer method: compare bands through bandwidth and MIMO. State the trade-off: low bands give coverage with limited rate, mid-bands balance rate and coverage, mmWave gives high capacity but short range.

Common mistake

Do not compare uplink and downlink only by antenna gain. UE transmit power is much lower than base-station transmit power, so uplink often fails first.

Figure 10.22 - Antenna gain and size benchmarking

Chapter 10 - Beamforming antenna gain



Screenshot 18. Figure 10.22 - Antenna gain and size benchmarking.

Explanation

This figure explains how 3.5 GHz massive MIMO partly compensates for the higher propagation loss of mid-band frequencies. As frequency increases, wavelength decreases, so more antenna elements can fit into the same physical area. A low-band passive antenna gives around 15 dBi, a 2 GHz passive antenna gives around 18 dBi, and a 3.5 GHz massive MIMO antenna can reach around 25 dBi with a similar or smaller frontal area. The extra 7-10 dB gain is important because 3.5 GHz propagation is worse than 1.8 or sub-1 GHz. However, higher gain comes from narrower beams, so the system needs beamforming to steer energy toward users. In exams, link this figure to both coverage and capacity: massive MIMO improves the link budget and spatial reuse, but uplink coverage can still be limited by UE power.

Important notes

Higher frequency allows smaller antenna elements and more elements in the same area. More elements enable higher gain and beamforming, not just a smaller box.

Exam answer pattern

Answer method: explain that higher frequency allows more antenna elements in a similar area, giving higher gain and narrower beams. Then link the result to beamforming and coverage compensation.

Common mistake

Do not say massive MIMO solves all coverage issues. Downlink improves strongly, but uplink may still need low-band support because the UE transmit power is limited.